

CS 441 Discrete Mathematics for CS
Lecture 9

Functions II

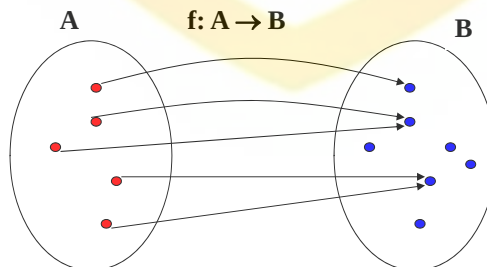
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Functions

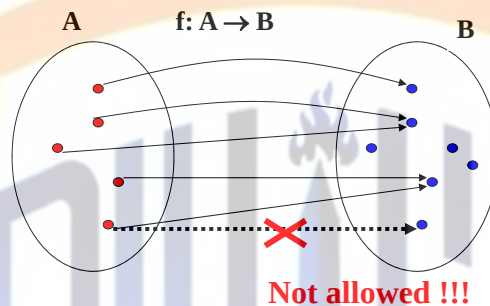
- **Definition:** Let A and B be two sets. A **function from A to B** , denoted $f: A \rightarrow B$, is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ to denote the assignment of b to an element a of A by the function f .



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Functions

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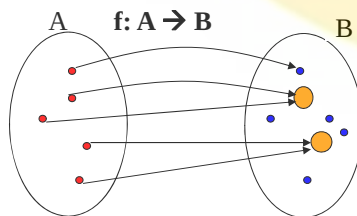
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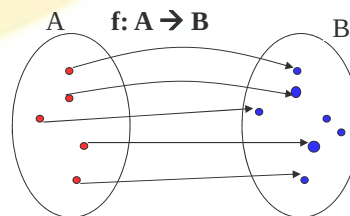
Injective function

Definition: A function f is said to be **one-to-one, or injective**, if and only if $f(x) = f(y)$ implies $x = y$ for all x, y in the domain of f . A function is said to be an **injection if it is one-to-one**.

Alternative: A function is one-to-one if and only if $f(x) \neq f(y)$, whenever $x \neq y$. This is the contrapositive of the definition.



Not injective function



Injective function

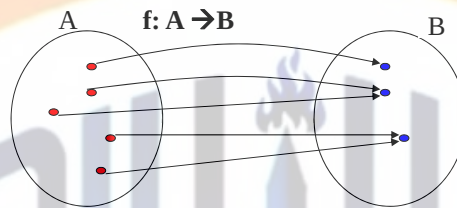
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Surjective function

Definition: A function f from A to B is called **onto, or surjective**, if and only if for every $b \in B$ there is an element $a \in A$ such that $f(a) = b$.

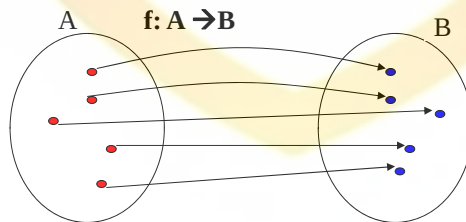
Alternative: all co-domain elements are covered



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Bijjective functions

Definition: A function f is called **a bijection** if it is **both one-to-one (injection) and onto (surjection)**.



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Bijjective functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
 - Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow b$
- Is f a bijection?
- ?

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Bijjective functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c\}$
 - Define f as
 - $1 \rightarrow c$
 - $2 \rightarrow a$
 - $3 \rightarrow b$
- Is f a bijection?
- **Yes.** It is both one-to-one and onto.

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Bijjective functions

Example 2:

- Define $g : W \rightarrow W$ (whole numbers), where
 $g(n) = \lfloor n/2 \rfloor$ (floor function).
 - $0 \rightarrow \lfloor 0/2 \rfloor = \lfloor 0 \rfloor = 0$
 - $1 \rightarrow \lfloor 1/2 \rfloor = \lfloor 1/2 \rfloor = 0$
 - $2 \rightarrow \lfloor 2/2 \rfloor = \lfloor 1 \rfloor = 1$
 - $3 \rightarrow \lfloor 3/2 \rfloor = \lfloor 3/2 \rfloor = 1$
 - ...
- Is g a bijection?

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Bijjective functions

Example 2:

- Define $g : W \rightarrow W$ (whole numbers), where
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 - $2 \rightarrow \lfloor 2/2 \rfloor = \lfloor 1 \rfloor = 1$
 - $3 \rightarrow \lfloor 3/2 \rfloor = \lfloor 3/2 \rfloor = 1$
 - ...
- Is g a bijection?
 - **No.** g is onto but not 1-1 ($g(0) = g(1) = 0$ however $0 \neq 1$).

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Bijjective functions

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Assume

→ A is finite and f is one-to-one (injective)

- Is f an onto function (surjection)?

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Bijjective functions

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Proof:

→ A is finite and f is one-to-one (injective)

- Is f an onto function (surjection)?
- **Yes.** Every element points to exactly one element. Injection assures they are different. So we have $|A|$ different elements A points to. Since $f: A \rightarrow A$ the co-domain is covered thus the function is also a surjection (and a bijection)

← A is finite and f is an onto function

- Is the function one-to-one?

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Bijjective functions

Theorem: Let f be a function $f: A \rightarrow A$ from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Proof:

→ **A is finite and f is one-to-one (injective)**

- Is f an onto function (surjection)?
- **Yes.** Every element points to exactly one element. Injection assures they are different. So we have $|A|$ different elements A points to. Since $f: A \rightarrow A$ the co-domain is covered thus the function is also a surjection (and a bijection)

← **A is finite and f is an onto function**

- Is the function one-to-one?
- **Yes.** Every element maps to exactly one element and all elements in A are covered. Thus the mapping must be one-to-one

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Bijjective functions

Theorem. Let f be a function from a set A to itself, where A is finite. Then f is one-to-one if and only if f is onto.

Please note the above is not true when A is an infinite set.

• **Example:**

- $f: \mathbb{Z} \rightarrow \mathbb{Z}$, where $f(z) = 2 * z$.
- f is one-to-one but not onto.
 - $1 \rightarrow 2$
 - $2 \rightarrow 4$
 - $3 \rightarrow 6$
- 3 has no pre-image.

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Functions on real numbers

Definition: Let f_1 and f_2 be functions from A to $\mathbf{R}$ (reals). Then $f_1 + f_2$ and $f_1 * f_2$ are also functions from A to $\mathbf{R}$ defined by

- $(f_1 + f_2)(x) = f_1(x) + f_2(x)$
- $(f_1 * f_2)(x) = f_1(x) * f_2(x).$

Examples:

- **Assume**

- $f_1(x) = x - 1$
- $f_2(x) = x^3 + 1$

then

- $(f_1 + f_2)(x) = x^3 + x$
- $(f_1 * f_2)(x) = x^4 - x^3 + x - 1.$

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Increasing and decreasing functions

Definition: A function f whose domain and codomain are subsets of real numbers is **strictly increasing** if $f(x) > f(y)$ whenever $x > y$ and x and y are in the domain of f . Similarly, f is called **strictly decreasing** if $f(x) < f(y)$ whenever $x > y$ and x and y are in the domain of f .

Example:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$. Is it increasing ?

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Increasing and decreasing functions

Definition: A function f whose domain and codomain are subsets of real numbers is **strictly increasing** if $f(x) > f(y)$ whenever $x > y$ and x and y are in the domain of f . Similarly, f is called **strictly decreasing** if $f(x) < f(y)$ whenever $x > y$ and x and y are in the domain of f .

Example:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$. Is it increasing ?

- **Proof .**

For $x > y$ holds $2x > 2y$ and subsequently $2x - 1 > 2y - 1$

Thus g is strictly increasing.

Increasing and decreasing functions

Definition: A function f whose domain and codomain are subsets of real numbers is **strictly increasing** if $f(x) > f(y)$ whenever $x > y$ and x and y are in the domain of f . Similarly, f is called **strictly decreasing** if $f(x) < f(y)$ whenever $x > y$ and x and y are in the domain of f .

Note: Strictly increasing and strictly decreasing functions are one-to-one.

Why?

Increasing and decreasing functions

Definition: A function f whose domain and codomain are subsets of real numbers is **strictly increasing** if $f(x) > f(y)$ whenever $x > y$ and x and y are in the domain of f . Similarly, f is called **strictly decreasing** if $f(x) < f(y)$ whenever $x > y$ and x and y are in the domain of f .

Note: Strictly increasing and strictly decreasing functions are one-to-one.

Why?

One-to-one function: A function is one-to-one if and only if $f(x) \neq f(y)$, whenever $x \neq y$.

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Identity function

Definition: Let A be a set. The **identity function** on A is the function $i_A: A \rightarrow A$ where $i_A(x) = x$.

Example:

- Let $A = \{1, 2, 3\}$

Then:

- $i_A(1) = ?$

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Identity function

Definition: Let A be a set. The **identity function** on A is the function $i_A: A \rightarrow A$ where $i_A(x) = x$.

Example:

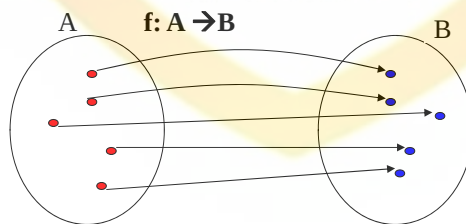
- Let $A = \{1, 2, 3\}$

Then:

- $i_A(1) = 1$
- $i_A(2) = 2$
- $i_A(3) = 3$.

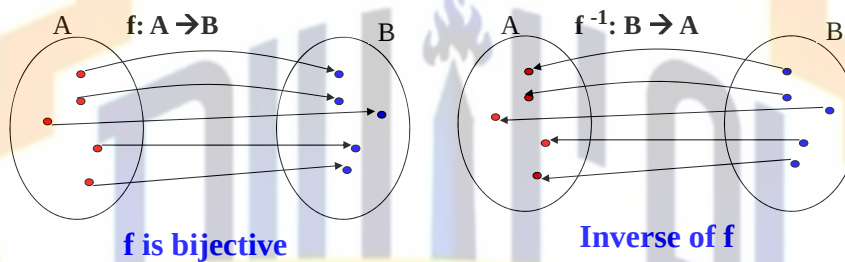
Bijjective functions

Definition: A function f is called **a bijection** if it is **both one-to-one and onto**.



Inverse functions

Definition: Let f be a **bijection** from set A to set B . The **inverse function of f** is the function that assigns to an element b from B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$, when $f(a) = b$. If the inverse function of f exists, f is called **invertible**.



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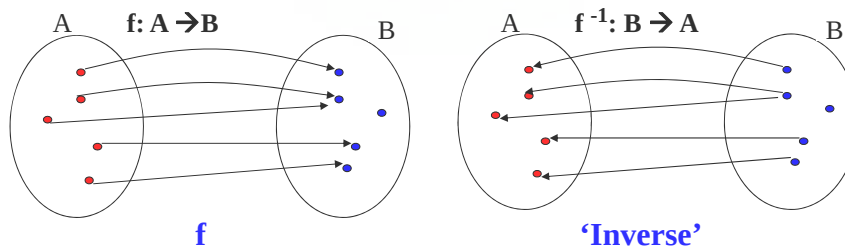
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Inverse functions

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is not one-to-one:

?



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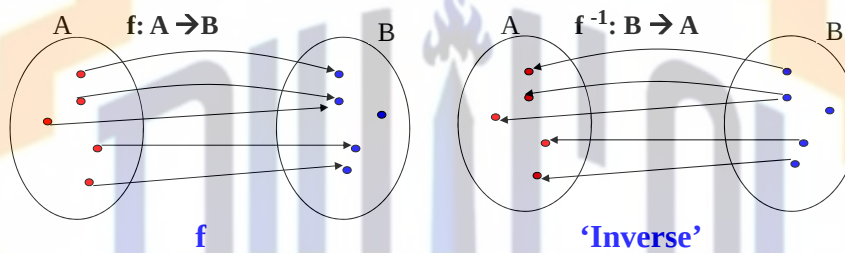
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Inverse functions

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is not one-to-one:

Inverse is not a function. One element of B is mapped to two different elements.



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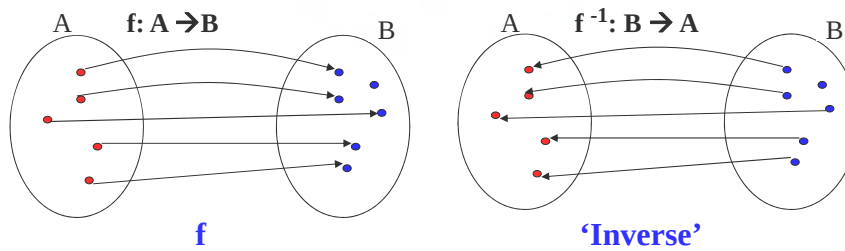
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Inverse functions

Note: if f is not a bijection then it is not possible to define the inverse function of f . **Why?**

Assume f is not onto:

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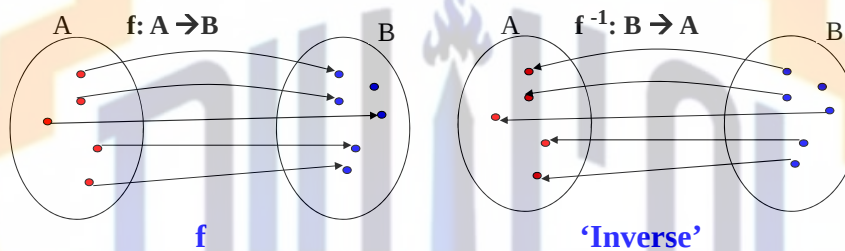
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Inverse functions

Note: if f is not a bijection then it is not possible to define the inverse function of f . Why?

Assume f is not onto:

Inverse is not a function. One element of B is not assigned any value in B .



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Inverse functions

Example 1:

- Let $A = \{1, 2, 3\}$ and i_A be the identity function

- | | | |
|---|--------------|-------------------|
| • | $i_A(1) = 1$ | $i_A^{-1}(1) = 1$ |
| • | $i_A(2) = 2$ | $i_A^{-1}(2) = 2$ |
| • | $i_A(3) = 3$ | $i_A^{-1}(3) = 3$ |

- Therefore, the inverse function of i_A is i_A .

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Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

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Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = ..$

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Inverse functions

Example 2:

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$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = 2*3 - 1 = 5$
- $g^{-1}(5) =$

Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = 2*3 - 1 = 5$
- $g^{-1}(5) = (5+1)/2 = 3$
- $g(10) =$

Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

Test the correctness of inverse:

- $g(3) = 2 \cdot 3 - 1 = 5$
- $g^{-1}(5) = (5+1)/2 = 3$
- $g(10) = 2 \cdot 10 - 1 = 19$
- $g^{-1}(19) =$

Inverse functions

Example 2:

- Let $g : \mathbf{R} \rightarrow \mathbf{R}$, where $g(x) = 2x - 1$.
- What is the inverse function g^{-1} ?

Approach to determine the inverse:

$$y = 2x - 1 \Rightarrow y + 1 = 2x$$

$$\Rightarrow (y+1)/2 = x$$

- Define $g^{-1}(y) = x = (y+1)/2$

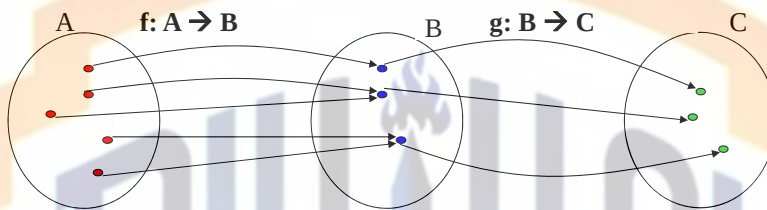
Test the correctness of inverse:

- $g(3) = 2 \cdot 3 - 1 = 5$
- $g^{-1}(5) = (5+1)/2 = 3$
- $g(10) = 2 \cdot 10 - 1 = 19$
- $g^{-1}(19) = (19+1)/2 = 10$.

Composition of functions

Definition: Let f be a function from set A to set B and let g be a function from set B to set C . The **composition of the functions g and f** , denoted by $g \circ f$ is defined by

- $(g \circ f)(a) = g(f(a))$.



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Composition of functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$g: A \rightarrow A,$	$f: A \rightarrow B$
$1 \rightarrow 3$	$1 \rightarrow b$
$2 \rightarrow 1$	$2 \rightarrow a$
$3 \rightarrow 2$	$3 \rightarrow d$

$f \circ g: A \rightarrow B:$

- $1 \rightarrow$

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Composition of functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$$g : A \rightarrow A, \quad f : A \rightarrow B$$

$$1 \rightarrow 3 \quad 1 \rightarrow b$$

$$2 \rightarrow 1 \quad 2 \rightarrow a$$

$$3 \rightarrow 2 \quad 3 \rightarrow d$$

$$f \circ g : A \rightarrow B:$$

- $1 \rightarrow d$
- $2 \rightarrow$

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Composition of functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$$g : A \rightarrow A, \quad f : A \rightarrow B$$

$$1 \rightarrow 3 \quad 1 \rightarrow b$$

$$2 \rightarrow 1 \quad 2 \rightarrow a$$

$$3 \rightarrow 2 \quad 3 \rightarrow d$$

$$f \circ g : A \rightarrow B:$$

- $1 \rightarrow d$
- $2 \rightarrow b$
- $3 \rightarrow$

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Composition of functions

Example 1:

- Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$

$$g : A \rightarrow A, \quad f: A \rightarrow B$$

$$1 \rightarrow 3 \quad 1 \rightarrow b$$

$$2 \rightarrow 1 \quad 2 \rightarrow a$$

$$3 \rightarrow 2 \quad 3 \rightarrow d$$

$$f \circ g : A \rightarrow B:$$

- $1 \rightarrow d$
- $2 \rightarrow b$
- $3 \rightarrow a$

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Composition of functions

Example 2:

- Let f and g be two functions from $\mathbb{Z}$ to $\mathbb{Z}$, where
- $f(x) = 2x$ and $g(x) = x^2$.
- $f \circ g : \mathbb{Z} \rightarrow \mathbb{Z}$
- $$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f(x^2) \\ &= 2(x^2) \end{aligned}$$
- $g \circ f : \mathbb{Z} \rightarrow \mathbb{Z}$
- $(g \circ f)(x) = ?$

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Composition of functions

Example 2:

- Let f and g be two functions from $\mathbb{Z}$ to $\mathbb{Z}$, where

- $f(x) = 2x$ and $g(x) = x^2$.

- $f \circ g : \mathbb{Z} \rightarrow \mathbb{Z}$

- $$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(x^2) \\ &= 2(x^2)\end{aligned}$$

- $g \circ f : \mathbb{Z} \rightarrow \mathbb{Z}$

- $$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(2x) \\ &= (2x)^2 \\ &= 4x^2\end{aligned}$$

Note that the order of
the function composition matters

Composition of functions

Example 3:

- $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$, for all x .

- Let $f : \mathbb{R} \rightarrow \mathbb{R}$, where $f(x) = 2x - 1$ and $f^{-1}(x) = (x+1)/2$.

- $$\begin{aligned}(f \circ f^{-1})(x) &= f(f^{-1}(x)) \\ &= f((x+1)/2) \\ &= 2((x+1)/2) - 1 \\ &= (x+1) - 1 \\ &= x\end{aligned}$$

Composition of functions

Example 3:

- $(f \circ f^{-1})(x) = x$ and $(f^{-1} \circ f)(x) = x$, for all x .
- Let $f : \mathbf{R} \rightarrow \mathbf{R}$, where $f(x) = 2x - 1$ and $f^{-1}(x) = (x+1)/2$.
- $(f \circ f^{-1})(x) = f(f^{-1}(x))$
 $= f((x+1)/2)$
 $= 2((x+1)/2) - 1$
 $= (x+1) - 1$
 $= x$
- $(f^{-1} \circ f)(x) = f^{-1}(f(x))$
 $= f^{-1}(2x - 1)$
 $= (2x)/2$
 $= x$

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Some functions

Definitions:

- The **floor function** assigns a real number x the largest integer that is less than or equal to x . The floor function is denoted by $\lfloor x \rfloor$.
- The **ceiling function** assigns to the real number x the smallest integer that is greater than or equal to x . The ceiling function is denoted by $\lceil x \rceil$.

Other important functions:

- Factorials: $n! = n(n-1) \dots 1$ such that $1! = 1$

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